

Practice Quiz 1 & Solution

Q1.

Give the result (in binary) of Verilog expression $A+(B|C)$ for the following inputs: $A=4'b1100$, $B=3'b100$ and $C=3'b010$. Assume A is a 4-bit wire and B and C are each 3-bit wires. Show your result using Verilog notation, such as $3b'101$.

****Solution****

```
4'b0010
//| = Bitwise OR
//+ = Addition
//See L4
```

Q2.

Give the result (in binary) of Verilog expression $(A==B)?B:C$ for the following inputs: $A=4'b1100$, $B=3'b100$ and $C=3'b010$. Assume A is a 4-bit wire and B and C are each 3-bit wires. Show your result using Verilog notation, such as $3'b101$.

****Solution****

```
3'b010
//Conditional Statement
```

Q3.

Give the result (in binary) of Verilog expression $\sim\&A$ for the following input: $A=4'b1100$. Assume A is a 4-bit wire. Show your result using Verilog notation, such as $3'b101$.

****Solution****

```
1'b1
//Reduction Operator ~&. Operates on all the bits within a word.
//Accepts a single word operand and produces a single bit as output.
//See L4.
```

Q4.

Write Verilog code that declares a 16-bit register, R_H1402 , and initially assigns it the hexadecimal value 1402.

****Solution****

```
reg [15:0] R_H1402;
initial R_H1402 = 16'h1402;
```

Q5.

What advantage is there to connecting ports by name, rather than position?

****Solution****

Many solutions accepted, but the primary reason is that it is more likely that wires will be connected to the correct port (i.e. it is harder to make a mistake).

Q6.

Write the hardware description of a 5-bit mod-22 counter. The suggested skeleton file has been written below:

```
module mod22Cntr (O, ck, rst);
    input ck, rst;
    output [4:0] O;
    //WRITE YOUR CODE HERE
```

```
endmodule
```

****Solution****

```
module mod22Cntr(O, ck, rst);
    input ck, rst;
    output [4:0] O;
    //WRITE YOUR CODE HERE
    reg [4:0] O;

    always @ (posedge ck) begin
        if (~rst) begin
            if (O == 5'b10101) begin
                O <= 5'b00000;
            end
            else begin
                O <= O+1;
            end
        end
    end
    always @ (posedge rst) begin
        O <= 5'b00000;
    end
endmodule
```
